



Full Length Article

Bridging gray-box modeling and machine learning: A digital twin approach to refrigeration system identification and predictive maintenance

Robert Pędzik , Mikołaj Suchon ^{*} , Tomasz Barszcz

Faculty of Mechanical Engineering and Robotics (WIMIR), AGH University of Krakow, Poland

ARTICLE INFO

Keywords:

Digital twins
Gray-box modeling
Refrigeration systems
Predictive maintenance
System Identification
Machine learning

ABSTRACT

This paper investigates the application of digital twins for refrigeration systems, focusing on system identification and predictive maintenance. The work explores the potential of thermodynamic modeling, using real physical parameters including thermal resistance, thermal capacity, and heat evaporation coefficient. The model's ability to detect changes in system parameters with minimal data is confirmed. The methodology includes a thermodynamic digital twin model developed in MATLAB, built for refrigeration system data. Traditional machine learning techniques were used for system identification and robust results validation. By embedding the compressor's hysteresis-based control logic, the closed-loop digital twin delivers accurate, high resolution real-time temperature predictions and paves the way for advanced disturbance detection and model-based control. Furthermore, the model successfully detected anomalies during an experiments with insulation degradation through cyclic door openings. These insights provide a foundation for advancing the use of digital twins in refrigeration system monitoring and maintenance

1. Introduction

Refrigeration systems play a critical role in preserving perishable goods such as food and pharmaceuticals. Ensuring their continuous operation requires effective predictive maintenance and system identification strategies. Traditionally, these objectives are met using statistical feature-based methods. However, a promising alternative lies in gray-box modeling, which enables the construction of digital twins that reflect the system's physical behavior using real-world parameters. System identification involves building mathematical models based on measured data to understand the system's dynamics, as described in [1], while predictive maintenance uses real-time data to predict when maintenance is needed, thereby preventing unexpected failures. Both are crucial for minimizing downtime, reducing energy consumption, and prolonging the equipment's lifespan. In experimental tests simulating reduced insulation via cyclic door openings, the model demonstrated its capability to detect anomalies corresponding to physical parameter changes.

By continuously simulating the physical system in real-time, digital twins can predict future behavior, diagnose issues before they occur, and optimize system performance. For refrigeration systems, this approach allows for a detailed understanding of system dynamics, helping

operators to predict failures, identify inefficiencies, and ensure smooth operation. The advantages of digital twins are vast, including real-time monitoring, more accurate predictive maintenance, and the ability to test system changes virtually before applying them to the physical system.

Machine learning enhances digital twin technology by providing advanced data analysis and pattern recognition capabilities. Through machine learning algorithms, a digital twin can continuously learn from new data, improving its accuracy in predicting system behavior and identifying potential issues. This synergy between machine learning and digital twins enables a more adaptive, intelligent system capable of self-optimization.

The objective of this paper is to explore the application of digital twin models in refrigeration systems, focusing on system identification and predictive maintenance. By developing a model in MATLAB and validating it with established machine learning tools, this research aims to assess the potential of the digital twin approach to enhance the efficiency and reliability of refrigeration systems across residential, commercial, and industrial environments.

1.1. State of the art

This work builds on early foundational studies in preventive

^{*} Corresponding author at: Jerozolimska 2a/49, 30-555, Krakow, Poland.

E-mail address: suchon@student.agh.edu.pl (M. Suchon).

Nomenclature

Model parameters

R	Thermal resistance of the cabinet walls (K / W)
C	Thermal capacitance of the cabinet contents (J / K)
G	Heat-removal (evaporation) coefficient of the compressor (W / K)

States and signals

T(t)	Internal cabinet temperature (°C)
Ta(t)	Ambient (room) temperature (°C)
u(t)	Compressor state indicator (binary: 1 = on, 0 = off)
M _r	Refrigerant mass (kg)
V _c (t)	Analogous “capacitor voltage” representing T(t) (V)
I(t)	Analogous “current” representing heat flow from evaporator (A)

Derived quantities

$\dot{V}_c = \frac{dV_c}{dt}$	Rate of change of capacitor voltage (V / s)
$\dot{Q} = I \cdot V$	Heat flow rate into the cabinet (W)
ΔT_{pp}	Peak-to-peak temperature swing per compressor cycle (K)
μ_{cycle}	Mean temperature per compressor cycle (K)
τ_{cycle}	Duration of one on–off compressor cycle (s)

Optimization and validation

P = [R, C, G]	Parameter vector to be estimated
RMSE	Root-mean-squared error between simulated and measured T(t) (K)
GA	Genetic-algorithm optimizer (MATLAB Global Optimization Toolbox)
chunk_k	kth continuous segment of stationary operation used for independent fitting
N_chunks	Total number of stationary segments (“chunks”) extracted

maintenance that focused on raw sensor data processing and generic anomaly detection. For example, the methodology presented in [2] demonstrates how careful data cleaning, feature extraction, and statistical filtering can reduce false positives in railway maintenance applications, yet provides limited insight into the physical causes of anomalies. To address the lack of standardized evaluation in temporal, unsupervised anomaly detection, [3] introduced an interval-based generalization of positive/negative instances and the Preceding Window ROC metric, explicitly rewarding early fault detection in predictive-maintenance scenarios. Their framework highlights the need for domain-tailored metrics when benchmarking algorithms on unlabeled time series. Additionally, [4] models the performance of a variable-speed compressor refrigeration system using artificial neural networks trained on limited experimental data, demonstrating accurate prediction of system behavior across different compressor frequencies and cooling loads.

Recent survey efforts have mapped the expanding landscape of anomaly detection and predictive maintenance within Industry 4.0 and compressor-based machines. The survey [5] reviews a spectrum of techniques—from statistical process control to deep-learning autoencoders—emphasizing the growing role of unsupervised representation learning and the persistent need for interpretability and resource efficiency in manufacturing environments. Complementing this, [6] categorizes tasks—fault detection (FD), fault prediction (FP), forecasting, and change-point detection (CPD)—and compares algorithm families such as LSTM, 1D-CNN, SVM, and tree-based models, while calling attention to the scarcity of public benchmarks and real-world datasets.

Building on these evaluations and surveys, several works have deployed data-driven machine learning for anomaly detection, albeit often at the cost of complex instrumentation or heavy computational demands. The 2025 study in [7] uses pressure (capacitive), current (Hall effect), voltage (inductive), and temperature (infrared, thermocouple) sensors with Arduino-based acquisition and Raspberry Pi processing to achieve high classification accuracy—but requires an elaborate hardware setup. Similarly, [8] first applies Isolation Forest to unlabelled IoT data (temperature, humidity, CO, LPG, smoke), generates pseudo-labels, and then trains Random Forest, MLP, and AdaBoost classifiers, yielding up to 99.93 % accuracy—though at the expense of large datasets and significant training time.

More recent refrigeration-specific frameworks seek to balance accuracy with practical deployability, [9] constructs virtual sensors from physical relationships, then applies tree-based algorithms (Random Forest, XGBoost, CatBoost, LightGBM) and SHAP for interpretability, achieving 99.48 % fault-classification accuracy using only pre-installed sensors. In parallel, [10] framework employs categorical gradient boosting regression on features such as coefficient of performance, subcooling, superheat, mass flow, and compression ratio, combined with advanced thresholding to detect both slow and catastrophic leaks—independently of refrigerant-level sensors—and flags slow leaks within an average of five days, substantially faster than technician inspection cycles.

In contrast, this work adopts a gray-box modeling approach inspired by [11], using measured data including internal and external fridge temperatures and compressor state. The foundation of this method was described in [12] with the aim of modeling the heat dynamics of buildings using thermodynamic analogies to electrical circuits, simulating systems with thermal resistance, thermal mass, and heat input sources—an approach applicable to refrigeration systems. Building on this, [11] applied the gray-box modeling technique to refrigeration units to estimate produce mass from system behavior, researching the method’s effectiveness in identifying physical parameters for classification. However, it also highlighted the need for further work in fault detection and abnormal operation tracking. Both studies rely on the parameter estimation framework from [13], which uses a maximum likelihood method with Kalman filtering based on stochastic differential equations. This paper extends the work in [11], re-implementing the model in MATLAB with a genetic algorithm for parameter optimization. It also introduces a new protocol for data processing and model validation, focusing on the digital twin’s utility for system identification and predictive maintenance. The contribution lies in bridging the heavily explored machine learning paradigm with gray-box modeling, leveraging industry-standard data processing for anomaly detection, robust optimization algorithms, and powerful dynamic models.

1.2. Description of the research object

Refrigeration cabinets (Fig. 1) and freezer cabinets (Fig. 2) are among the most commonly used professional refrigeration devices in the food trade, designed for the storage or display of food products. Depending on the operating temperature range, refrigeration devices intended for beverages and products requiring cooling are distinguished from freezer devices designed for the storage of frozen semi-finished products, vegetables, and fruits.

These devices are equipped with a refrigerated chamber with doors allowing access to the stored goods, and with either an integrated refrigeration system (Fig. 3) or a partial system connected to an external condensing unit. This system includes two heat exchangers connected by a pipeline — the Evaporator (1) and the Condenser (2) — as well as a Compressor (3) that increases the refrigerant pressure, and an Expansion Valve (4) that reduces it. The refrigerant serves as a medium that transports thermal energy from a lower-temperature environment (Tk – inside the chamber) to a higher-temperature environment (To – the surroundings).



Fig. 1. (left): Refrigerated cabinet by ES System K Ltd.



Fig. 2. (right): Freezer cabinet by ES System K Ltd.

The operating principle of the vapor-compression system is based on the laws of thermodynamics. According to the first law of thermodynamics (the law of conservation of energy), the system does not produce cold, but instead transfers heat from one area to another at the expense of the compressor's work. The second law of thermodynamics determines the direction of energy flow — heat cannot spontaneously flow from a colder medium to a warmer one; therefore, mechanical work must be supplied during the compression process.

In practice, the refrigeration cycle consists of four basic processes:

- Evaporation – in the evaporator, the refrigerant evaporates, absorbing heat from inside the chamber.
- Compression – the compressor raises the pressure and temperature of the refrigerant vapor.
- Condensation – in the condenser, the refrigerant releases heat to the surroundings and condenses.
- Expansion – the expansion valve lowers the refrigerant's pressure and temperature, allowing it to evaporate again.

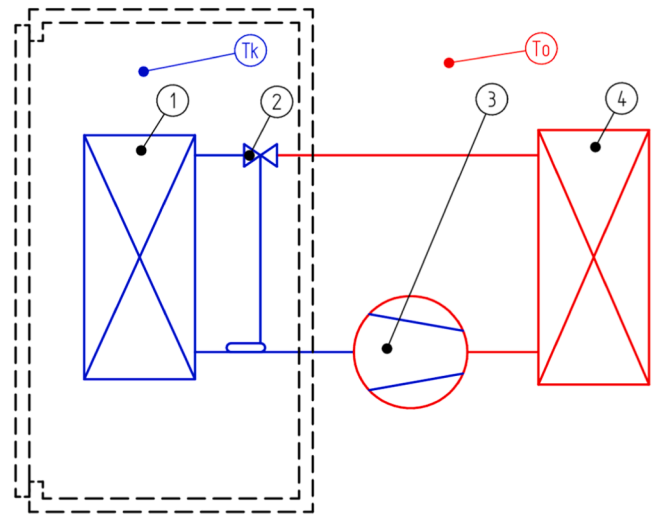


Fig. 3. Refrigeration system diagram: 1-Evaporator, 2-Expansion valve, 3-Compressor, 4-Condenser, Tk-Temperature of the refrigerated chamber, To-Ambient temperature.

The transfer of thermal energy in this cycle is based on phase-change phenomena — evaporation and condensation — as well as on the Joule–Thomson effect, which causes the refrigerant to cool during expansion. The behavior of the refrigerant in different phases is described by the ideal gas law ($pV = nRT$) and the Clapeyron–Clausius equation, which defines the relationship between temperature and pressure during phase-change processes.

Thus, the vapor-compression refrigeration system realizes a reversed Carnot cycle, in which mechanical energy supplied by the compressor enables the transfer of heat from a colder area to a warmer one. In practice, this means that the device operates in accordance with the laws of thermodynamics, utilizing the phase transitions of the refrigerant to maintain the required temperature parameters of food products.

2. Digital twin modeling

2.1. Assumptions of the electrical model

Using the standard thermal–electrical analogy (temperature $\equiv$ voltage, heat flow $\equiv$ current, thermal resistance $\equiv$ resistor, heat capacity $\equiv$ capacitor), our per-cabinet/evaporator grey-box model is best interpreted as follows: the cabinet (air + goods + inner surfaces) effective heat capacity is represented by a capacitor (storing thermal energy). Heat loss to the surroundings is represented by a resistor whose value corresponds to the thermal resistance of the enclosure. The evaporator appears as a heat-exchange element (a heat sink) whose magnitude depends on refrigerant mass and an evaporator heat-transfer coefficient (or UA); in the analogy this can be written as a time-varying current out of the cabinet capacitor, but the evaporator itself is not modelled as a storage capacitance. The upstream compressor/condenser/expansion components are outside the cabinet model: their effect on the cabinet is captured through the mass-flow input (or equivalent heat flux) to the evaporator, which is often intermittent (hysteresis/modulation of the valve). This matches the modelling choices and equations used in the referenced grey-box literature [11, 12]. The resulting electrical model enables analysis of the dynamic behavior of the refrigeration system in a manner analogous to a forced RC circuit, allowing the simulation of thermal energy storage and exchange processes over time.

2.2. Digital twin model

The model depicted in Fig. 4 simulates the refrigerator's internal temperature based on a set of physical parameters: thermal resistance, thermal capacitance, and the compressor's heat removal coefficient. The strength of this model lies in its broad applicability to most refrigeration systems comprising a thermally insulated chamber, an ambient-temperature environment, and a heat exchanger. The inputs to the model are ambient temperature and compressor state over time, and by comparing the simulated outputs, example of which is present in Fig. 5, with actual temperature measurements, system parameters can be estimated through optimization techniques. In this implementation, a genetic algorithm from MATLAB's Global Optimization Toolbox [14] is employed for parameter fitting.

Deriving the differential equations for the system, we obtain:

$$\frac{dq_c(t)}{dt} = -\frac{1}{RC}q_c(t) + I(t) + \frac{1}{R}T_a$$

$$V_c(t) = \frac{1}{C}q_c(t)$$

- V_c represents the voltage across a capacitor, indicating the temperature inside the fridge.
- R represents the thermal resistance of the fridge walls.
- I represents the current of energy flow from the evaporator (related to the equation in Fig. 1).
- C represents the thermal capacitance of the inside of the fridge, including air and produce.
- $\frac{dq_c(t)}{dt}$ represents the rate of change of charge of the capacitor over time. Charge being analogous to the temperature inside the cabinet.
- T_a represents the ambient room temperature.

From the differential equations, we can then model the system's behavior by deriving the system's state-space equations:

$$A = -\frac{1}{c * R}$$

$$B = \left[G \frac{1}{R} \right]$$

$$C = \frac{1}{c}$$

$$D = [0 \ 0]$$

where:

- A is the state matrix.
- B is the input matrix.
- C is the output matrix.
- D is the feedforward matrix.

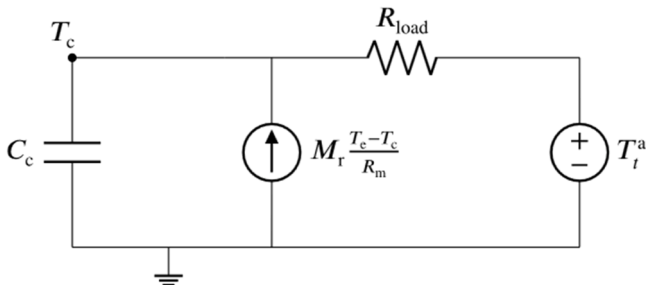


Fig. 4. RC-diagram of the model. The refrigerant mass state, M_r , is included in the heat source illustrating the heat transfer to the cabinet. Source: [11].

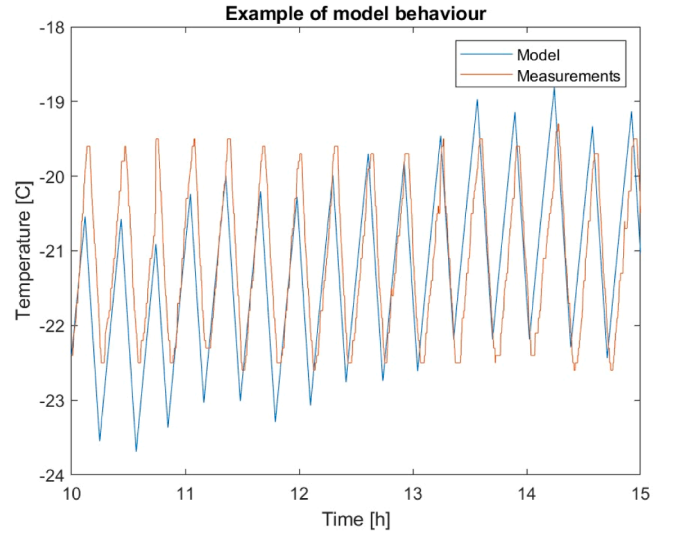


Fig. 5. Example of the model simulating the temperature inside a fridge. The data was collected over 5 h of normal freezer operation and includes ambient temperature T_a and compressor state as the model's inputs. We estimate R , C , and G .

- c represents the thermal capacitance of the inside of the fridge, including air and produce.
- G is the heat evaporation factor, Section 5.3 in [15].

2.3. Anomaly detection

To ensure meaningful parameter identification, the input data must accurately reflect the system's stationary behavior. Data segments influenced by disturbances—such as prolonged door openings or defrosting cycles—are excluded, as they temporarily distort the apparent physical parameters (e.g., reducing perceived thermal resistance). Put simply, parameter identification relies exclusively on measurements recorded during steady-state operation. Data preprocessing is therefore essential to remove these non-stationary intervals. This is done by sampling the dataset into cycles (defined between compressor on-off events) and calculating traditional statistical features such as cycle length, mean temperature per cycle, and peak-to-peak variation of temperature per cycle to identify and discard anomalies. Fig. 6 presents the result of a statistical feature-driven algorithm for anomaly detection. In this specific example peak-to-peak was used as the discriminatory feature. Data processing for training comprised removal of measurement errors (replaced by linear interpolation), detection of compressor cycles (as defined above), extraction of statistical features per cycle, selection of the most discriminating feature, and application of MATLAB's `isoutlier()` function—flagging observations more than three scaled median absolute deviations from the median as anomalies. Refer to the anomaly detection algorithm in ch.6.3 in [15] for more information on dataset preparation for parametric identification using Machine Learning in refrigeration systems.

2.4. Methodology

As a result of filtering, the dataset is divided into multiple continuous segments (termed “chunks”), each representing a period of stable operation. The model is trained independently on each chunk, yielding multiple sets of identified parameters. This chunk-based training not only enhances generalization but also increases statistical robustness, enabling effective exploration of the parameter space. In the case of a three-parameter model, results form a curve in 3D space. If one parameter is known or measured independently, the remaining two can be inferred with good accuracy. Figures 4.1–4.3 depict the curve of

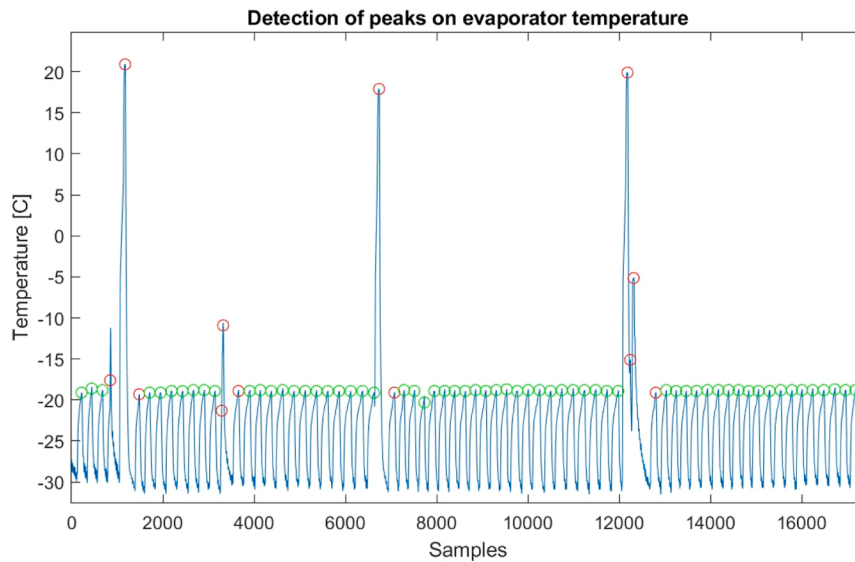


Fig. 6. Result of anomaly detection protocol.

results and demonstrate the possibility of parametric identification given that one of three parameters is known.

2.5. Parametric identifiability

Let R, C, G be real positive parameters. Consider the single-state LTI model

$$\dot{x}(t) = -\frac{1}{RC}x(t) + Gu_1(t) + \frac{1}{R}u_2(t), \quad y(t) = \frac{1}{C}x(t),$$

Its transfer functions from the two inputs to y are

$$H_1(s) = \frac{G/C}{s + \alpha}, H_2(s) = \frac{\alpha}{s + \alpha}, \quad \text{where } \alpha := \frac{1}{RC}.$$

From sufficiently rich I/O data the rational functions H_1, H_2 (hence the pole α and the residue $r_1 := G/C$) are uniquely determined. Thus the measured quantities are

$$\alpha = \frac{1}{RC}, \quad r_1 = \frac{G}{C}.$$

Assume one parameter is known; solve the two algebraic equations above for the remaining two parameters.

1. R known: $C = \frac{1}{R\alpha}$ and $G = r_1 C = \frac{r_1}{R\alpha}$.
2. C known: $R = \frac{1}{\alpha C}$ and $G = r_1 C$.
3. G known (and $r_1 \neq 0$): $C = \frac{G}{r_1}$ and $R = \frac{1}{\alpha C} = \frac{r_1}{\alpha G}$.

Each case yields a unique solution (except the degenerate case $r_1 = 0$ when $G = 0$, which prevents recovering C from r_1). Therefore, under nondegeneracy and with sufficiently exciting inputs, knowing any one of R, C, G makes the remaining two parameters globally (algebraically) identifiable.

2.6. Parametric uncertainty

The identifiability proof was augmented with a quantitative uncertainty analysis based on the Fisher Information Matrix. First, from the end-to-end residuals we obtain

$$\hat{\sigma}_y^2 = 2.23 \text{ } ^\circ\text{C},$$

then propagate sensitivities via forward-Euler to form $F = J^T J$ and estimate the full parameter covariance as $\text{Cov}(\theta) = \hat{\sigma}_y^2 F^{-1}$.

Numerically this yields

$$\text{Cov}(\theta) = \begin{bmatrix} 0.021 & 1.1300 & -0.4864 \\ 1.1300 & 59.2267 & -25.4934 \\ -0.4864 & -25.4934 & 10.9738 \end{bmatrix}$$

with one eigenvalue effectively zero ($\lambda_1 \approx 0$), confirming that (G, c, R) lie on a one-dimensional manifold in 3D. The one-sigma standard deviations of the estimates are:

$$\sigma_G = 0.1468 \quad \sigma_c = 7.6959 \quad \sigma_R = 3.3127$$

Thus confirming the experiment observation (Figs. 7.1-7.3) that c is the least precisely determined parameter.

To visualize parameter correlations and the effect of fixing one parameter, Figs. 8.1-8.3 show the 95 % confidence ellipses in the two-parameter planes obtained by conditioning $\text{Cov}(\theta)$ on G, c or R , respectively. Fixing G yields the smallest ellipse area (i.e. the tightest confidence regions for c and R), fixing R is next best, and fixing c leads to the largest uncertainty in the remaining two parameters. These ellipses both illustrate the strong linear correlation among (G, c, R) and quantitatively demonstrate that, once any single parameter is known, the other two become well-identified with readily computed confidence intervals.

2.7. Optimization algorithm

The genetic algorithm (GA) optimization in this study was performed using MATLAB's built-in `ga` function. The decision variables were the three model parameters (thermal resistance, thermal capacity, and evaporation coefficient).

Hyperparameters used:

- **Population size:** 50
- **Selection function:** stochastic uniform
- **Crossover fraction:** 0.8 (scattered crossover)
- **Mutation function:** adaptive feasible
- **Stopping criteria:** 100 generations or stall limit of 50 generations
- **Constraint handling:** default penalty method

The optimization was initialized using MATLAB's pseudo-random number generator. In our initial experiments, we used the default random seed. Although the GA is stochastic, the optimization was repeated hundreds of times to yield one results curve, so the inherent randomness was already sampled extensively, and re-running with

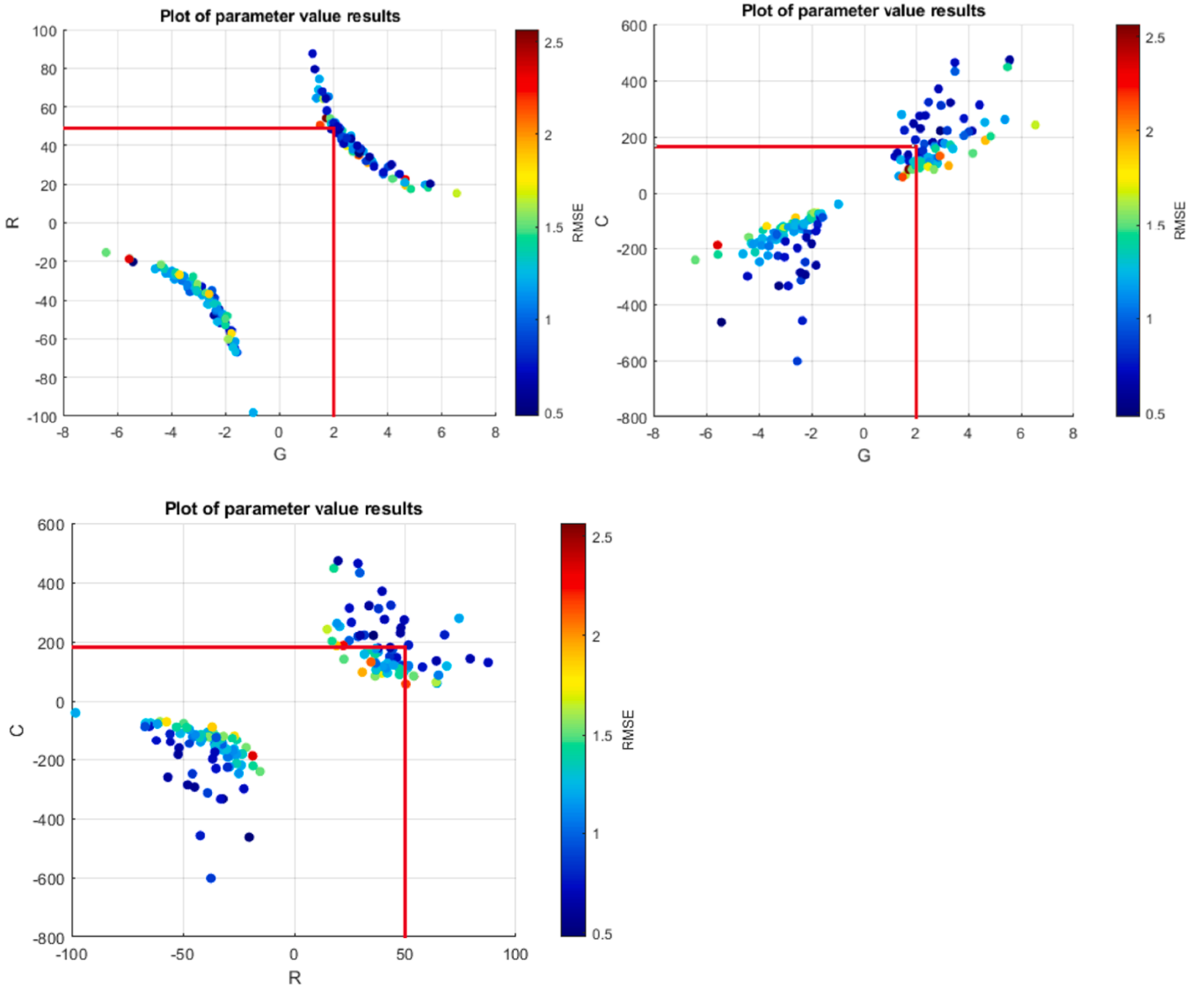


Fig. 7.1. (G-R plane), Figure 7.2 (G-C plane) and Figure 7.3 (R-C plane). A conceptual example of a possible parametric identification technique (imagining we know $G = 2$, which implies $R = 50$ and $C = 190$) overlaid on data from the above-mentioned experiment with model RMSE accounted for.

different seeds would not alter the reported results.

MATLAB's built-in Nelder–Mead solver (`fminsearch`) was also tested on this problem, using both a random start point and the best solution from a one-plus-one run. In both cases it achieved only a marginal RMSE drop (from 0.8007 to 0.7917) and produced essentially the same fit, indicating it was trapped in a nearby local minimum. This confirms that simple direct-search methods lack the global exploration needed for our highly multimodal cost surface, whereas our Genetic-Algorithm-based approach more reliably finds better optima. Section 7.2 in [15].

3. Real-time temperature simulations

To evaluate the model's capability to simulate real-time temperature, each parameter set, obtained from training on a data chunk, is tested against all other chunks from the training set by computing the RMSE (root mean squared error) between simulated and measured temperatures. The parameter set with the lowest average RMSE across all chunks in the training set, excluding the chunk it was trained on, is

selected as the best representation of the system. This protocol is drawn out in Fig. 9. This parameter set is then tested on data from the testing set ensuring the model was tested on new, unseen data that was not used in fitting.

While the model excels at parametric identification, its performance in continuous-time simulation is limited when compressor logic is omitted. Since the compressor input in the simulation is based on real-time data rather than modeled behavior, any mismatch in the system's simulated temperature introduces errors that accumulate over successive cycles. This leads to significant divergence between simulated and actual temperatures over time. To improve time-domain prediction, models need to replicate the compressor control logic (e.g., on/off decisions based on internal temperature thresholds). This was achieved by simulating the temperature dynamics of a refrigerator using a discrete-time state-space model with hysteresis control, updating the system state at each time step via Euler integration and outputting the time evolution of states, temperatures, and control inputs (Fig. 10). Please note that this simulation relies on the parameter set identified from the

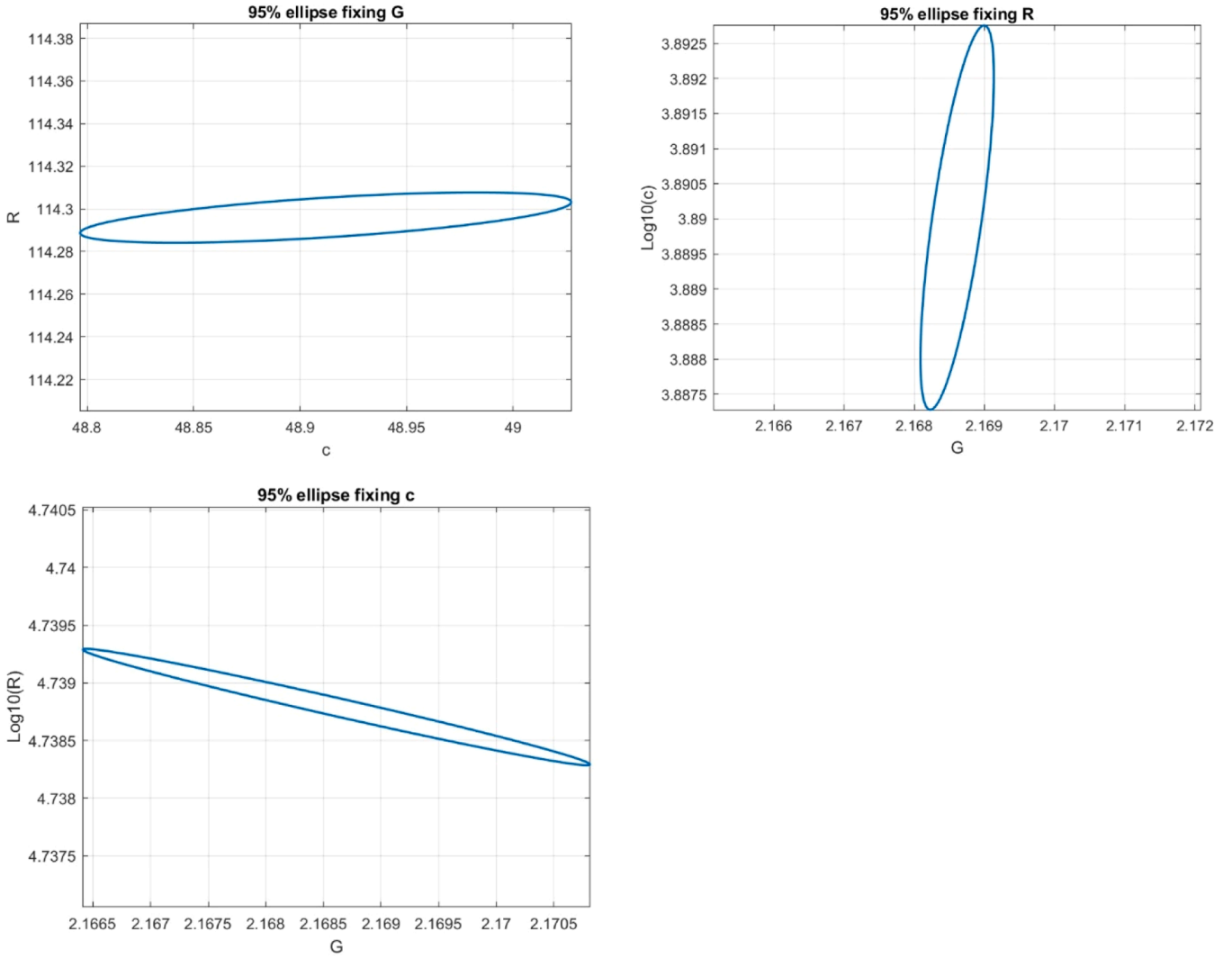


Fig. 8.1–8.3. 95 % confidence ellipses in the two-parameter planes obtained by conditioning $\text{Cov}(\theta)$ on G , c or R , respectively.

training dataset and is validated against independent test data with no leakage between sets. These results highlight the model's potential for short-term temperature prediction and applications such as detecting disturbances (e.g., door openings), although further research is still required.

An initial hypothesis held that, by operating directly on temperature in time rather than extracting cycle-based statistical features, the model would deliver both high temporal resolution and strong predictive performance. In contrast, statistical features force a compromise: finer resolution degrades feature quality, while more robust features require coarser time steps. In practice, however, the open-loop implementation—replaying real compressor on/off signals instead of generating them—proved unable to sustain accurate time-domain forecasts. Although the model starts each simulation with the correct initial state, any end-of-cycle temperature mismatch carries over and amplifies in subsequent cycles, so long-run temperature trajectories drift markedly from the real data. This is why a closed-loop implementation, as shown in Fig. 10, is required for accurate time domain predictions.

4. Results

Nevertheless, the model proves highly effective in identifying physical parameters. This is demonstrated through an experiment where a controlled load was placed inside a refrigeration system at temperature approximately -18°C and the door was periodically opened for 6 s every 10 min. This period was deliberately chosen to match the compressor cycle length so that repeated door openings produce a quasi-steady infiltration load, emulating the thermal isolation fault of a deteriorated gasket rather than a transient structural change. The goal was to evaluate whether the digital twin model could identify distinct sets of physical parameters corresponding to two stationary periods with different real-world conditions: one with standard operation and one influenced by frequent door openings. The data was segmented using a feature-based protocol described in [15]. This process yielded two distinct datasets suitable for parameter identification and model validation. Using this segmented data, 100 models were trained per dataset. Each model was trained proportionally to the duration of its data chunk. The model successfully detected the difference and correctly attributed

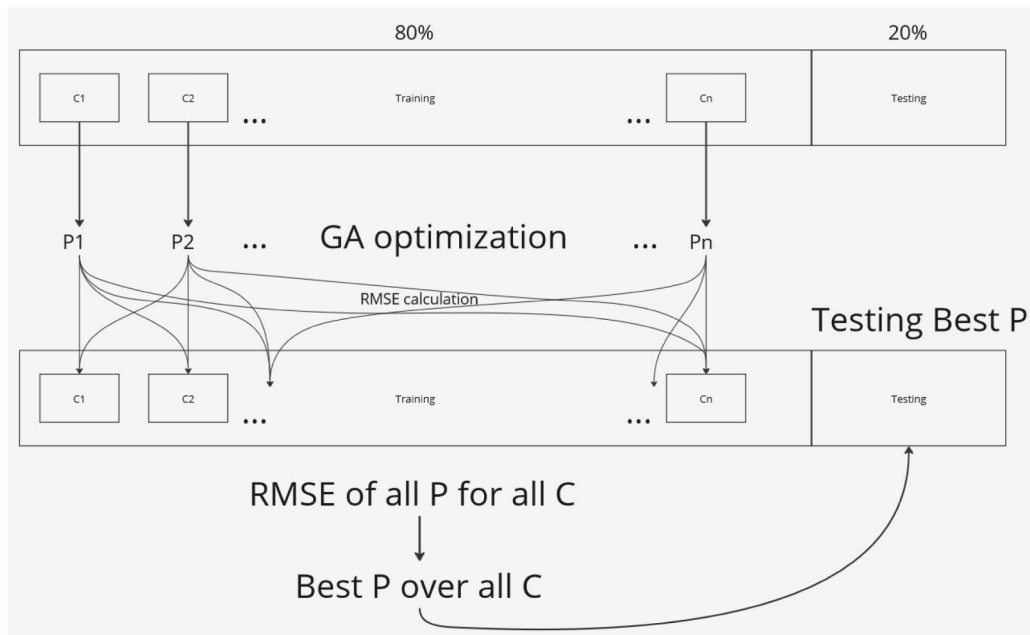


Fig. 9. Pictogram of data usage.

C - chunk of continuous stationary periodic signal

P - parameter set (G, R, C) resulting from GA optimization.

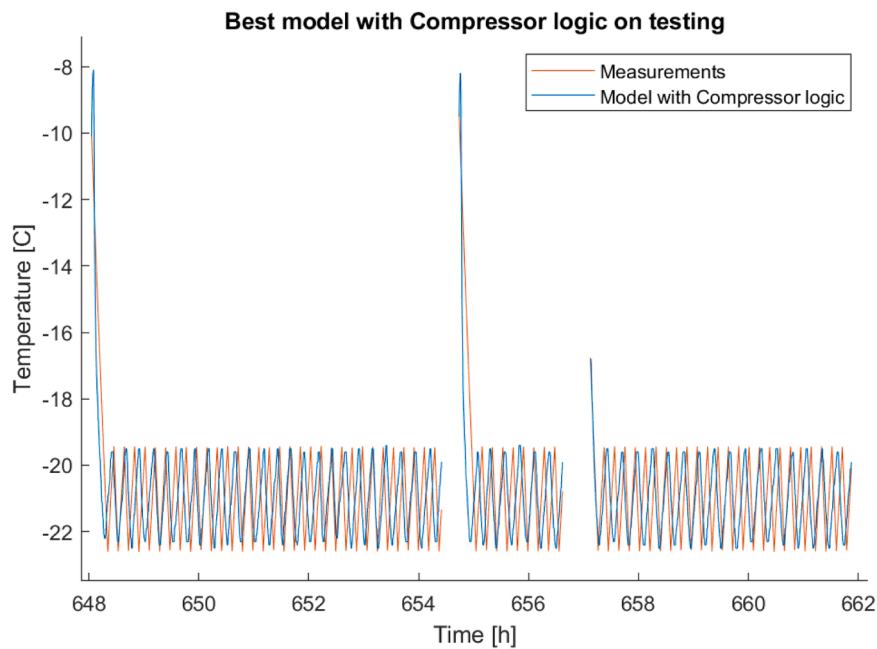


Fig. 10. Dynamic temperature simulation of a discrete-time state-space model with hysteresis control using the parameter set identified from the training dataset compared against independent test data. Visible on Figure is part of the test data set. The test data set was over 100 h of continuous operation. Model simulation had similar performance on all test data.

it to a decrease in thermal resistance, confirming its sensitivity to real physical changes. Fig. 11 showcases that the results are clearly separable in the G-R plane while results on the G-C plane (Fig. 12) perfectly overlap, implying that only the thermal resistance parameter (R) changed by decreasing, which aligns precisely with the experimental conditions. The experiment was replicated on a separate refrigeration system maintained at approximately 1 °C. Results visible on Figs. 13 and Fig. 14 confirm the hypothesis regarding the model's sensitivity to physical changes. The model's ability to accurately differentiate between thermal resistance (R), thermal capacity of the fridge's contents

(C), and the heat evaporation coefficient (G) is therefore proven.

4.1. Quantitative assessment of parameter shifts

To demonstrate statistical robustness and repeatability, paired comparisons of the identified parameters were performed for both Experiment 1 ($n = 100$) and Experiment 2 ($n = 98$). For each parameter we report the mean $\pm$ SD under "normal" versus "disturbed" conditions, the absolute and relative shift Δ , the 95 % confidence interval on Δ , Cohen's d for effect size, and the two-tailed paired t -test result.

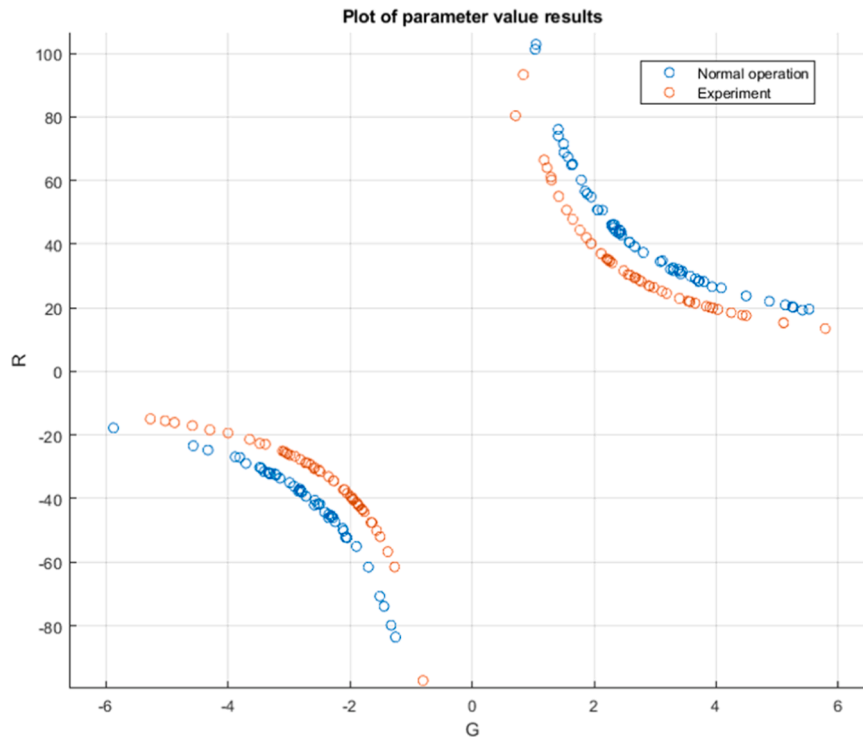


Fig. 11. (left): Plot of the result of training 100 models on normal operation data and data during the experiment of periodic door opening for a refrigeration system at temperature approximately -18°C . 2D view (G-R plane) of a 3D space.

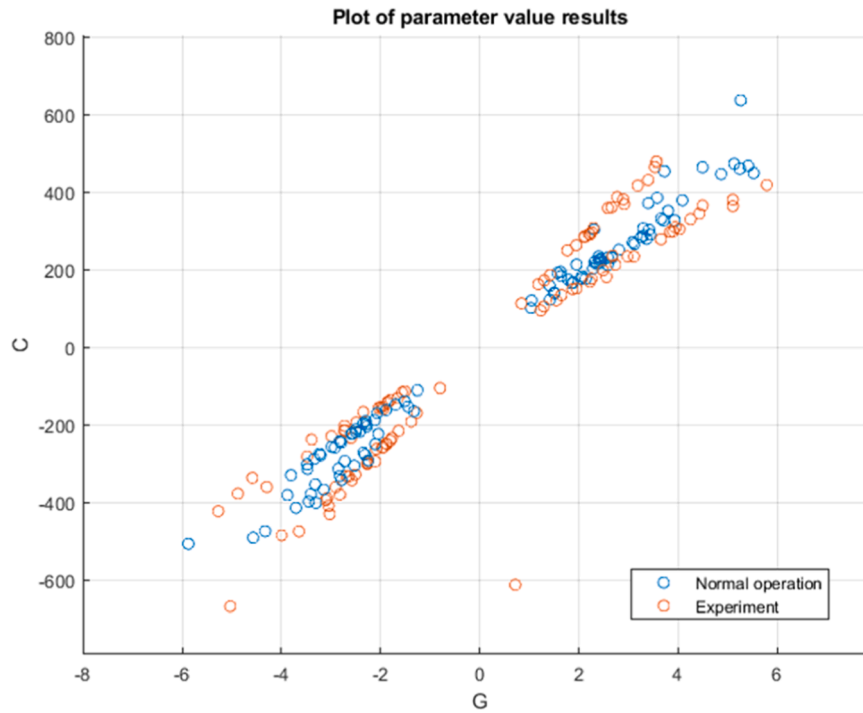


Fig. 12. (right): Plot of the result of training 100 models on normal operation data and data during the experiment of periodic door opening for a refrigeration system at temperature approximately -18°C . 2D view (G-C plane) of a 3D space.

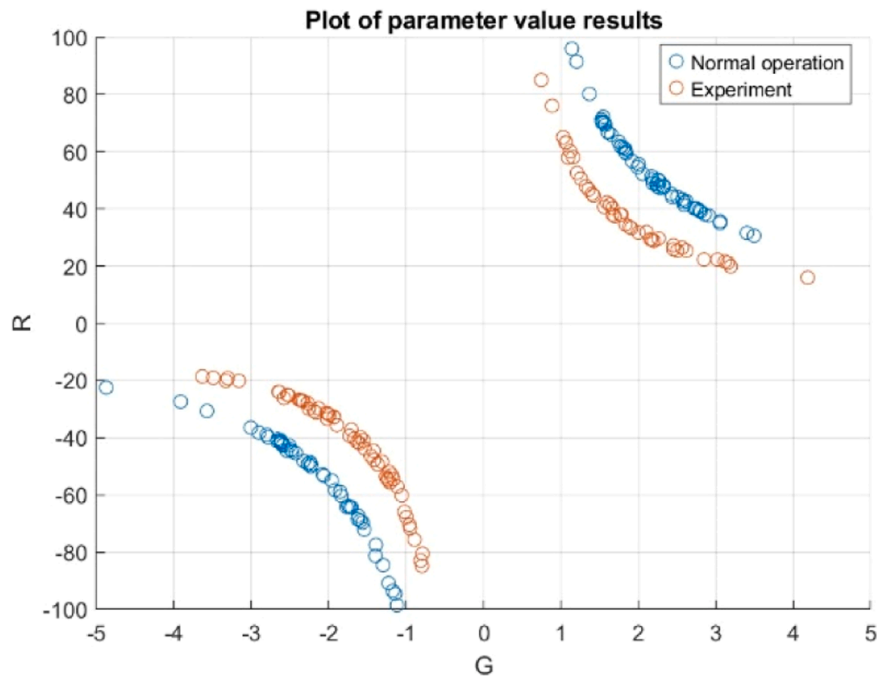


Fig. 13. (left): Plot of the result of training 100 models on normal operation data and data during the experiment of periodic door opening for a refrigeration system maintained at approximately 1 °C. 2D view (G-R plane) of a 3D space.



Fig. 14. (right): Plot of the result of training 100 models on normal operation data and data during the experiment of periodic door opening for a refrigeration system maintained at approximately 1 °C. 2D view (G-C plane) of a 3D space.

Experiment 1 (−18 °C system, n = 100)							
Var	Mean Normal ± SD	Mean Disturbed ± SD	Δ (abs) [%]	d	95 % CI	t (df)	p
G	2.84 ± 1.04	2.73 ± 1.23	−0.12 (−4.2 %)	−0.07	[−0.45, 0.21]	−0.70 (99)	0.484
R	42.63 ± 16.82	34.43 ± 15.63	−8.20 (−19.2 %)	−0.36	[−12.78, −3.62]	−3.54 (99)	0.000619
C	274.58 ± 103.48	282.37 ± 119.84	+7.79 (+2.8 %)	+0.05	[−26.19, 41.76]	+0.45 (99)	0.652

G: No significant change (negligible effect size).
R: Statistically significant, medium-sized decrease.
C: No significant change.

Experiment 2 (≈ 1 °C system, n = 98)							
Var	Mean Normal ± SD	"Mean Disturbed ± SD"	Δ (abs) [%]	d	95 % CI	t (df)	p
G	2.20 ± 0.65	1.84 ± 0.71	−0.36 (−16.3 %)	−0.38	[−0.54, −0.17]	−3.78 (97)	0.000268
R	54.42 ± 16.38	41.32 ± 16.64	−13.10 (−24.1 %)	−0.61	[−17.44, −8.76]	−5.96 (97)	4.07 × 10 ^{−8}
C	319.54 ± 98.28	283.15 ± 140.07	−36.40 (−11.4 %)	−0.21	[−71.40, −1.40]	−2.05 (97)	0.0427

G: Moderate-sized, significant decrease.
R: Large-sized, highly significant reduction.
C: Small effect, borderline significance.

Across two independent systems, only the thermal resistance R shows a consistent, medium-to-large and highly significant decrease under the door-opening disturbance ($p \ll 0.001$ in both cases), whereas G and C remain essentially unchanged or only weakly affected. This quantitative evidence confirms that the digital-twin identification framework reliably isolates changes in R, providing strong statistical support beyond anecdotal separation in scatter plots.

5. Conclusions

The final hypothesis addresses the advantages and limitations of using thermodynamic modeling for deriving digital twins for system parametric identification and anomaly prediction. The model operates on interpretable parameters such as thermal resistance, thermal capacity, and heat evaporation coefficient, offering significant benefits for fault prediction and maintenance. The use of non-abstract parameters enables empirical validation and separate measurements. Additionally, the model’s ability to capture the system’s natural behavior, with minimal data requirements, for parametric identification has been confirmed through testing and case studies. Not requiring much data, the model was able to identify two different instances of real parameters and correctly identify which parameters changed by how much. This experiment was repeated on two different systems, a refrigerator unit and a freezer unit.

To facilitate accurate temporal simulations, the refrigerator’s compressor logic was directly embedded into the model. By reformulating the continuous-time state-space equations in discrete time and adding hysteresis control—updating temperatures via Euler integration, comparing them to on/off thresholds, and switching the compressor accordingly—we close the loop on prediction. Using the same parameter set identified on the training data and validated against an independent test set, this closed-loop model tracks actual temperatures closely over time. While the original approach remains excellent for parametric identification, incorporating the compressor’s on/off logic is essential for reliable, high-resolution temperature prediction and lays the groundwork for applications such as disturbance detection and model-based control.

In addition to developing and implementing the model, an architecture integrating MATLAB genetic algorithms for parameter optimization was established. A comprehensive protocol for data processing was documented, encompassing data interpretation, error correction, and anomaly detection based on statistical features. Incorporating machine learning and heuristic concepts, the protocol ensures reliable

results through structured data preparation, feature-based identification, and bootstrap aggregation. The dataset division into training and testing sets follows a novel approach designed to maximize generalization and validation.

In conclusion, this approach offers a robust method for identifying physical parameters in refrigeration systems using gray-box modeling and digital twins. While the open-loop model remains unsuitable for accurate time-based simulation, incorporating the compressor’s on/off control logic enables reliable, high-resolution temperature prediction and lays the groundwork for predictive maintenance and model-based control.

CRedit authorship contribution statement

Robert Pędzik: Supervision, Data curation. **Mikołaj Suchoń:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Formal analysis. **Tomasz Barszcz:** Writing – review & editing, Validation, Supervision, Methodology, Investigation, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Tomasz Barszcz reports a relationship with AMC Vibro that includes: board membership. Robert Pedzik reports a relationship with ES Systems K that includes: employment. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

[1] O. Laguerre, Heat transfer and air flow in a domestic refrigerator. *Mathematical Modeling of Food Processing*, CRC Press, 2011, pp. 453–482.
[2] J. Rabatel, S. Bringay, P. Poncelet, Anomaly detection in monitoring sensor data for preventive maintenance, *Expert. Syst. Appl.* 38 (6) (2011) 7003–7015.
[3] J. Carrasco, D. López, I. Aguilera-Martos, D. García-Gil, I. Markova, M. García-Barzana, M. Arias-Rodil, J. Luengo, F. Herrera, Anomaly detection in predictive

- maintenance: a new evaluation framework for temporal unsupervised anomaly detection algorithms, *Neurocomputing*. 462 (2021) 440–452.
- [4] Ö. Kizilkan, Thermodynamic analysis of variable speed refrigeration system using artificial neural networks, *Expert. Syst. Appl.* 38 (2011) 11686–11692.
 - [5] P. Kamat, R. Sugandhi, Anomaly detection for predictive maintenance in Industry 4.0 – A survey, in: *E3S Web of Conferences*, EDP Sciences, 2020, 170.
 - [6] F.P. Forbicini, N.O. Vago, P. Fraternali, Time series analysis in compressor-based machines: a survey, *Neural Comput. Appl.* 37 (17) (2025) 11001–11038.
 - [7] A. Quispe-Astorga, R.J. Coaquira-Castillo, L.W. Utrilla Mego, J.C. Herrera-Levano, Y. Concha-Ramos, E.J. Sacoto-Cabrera, E. Moreno-Cardenas, Data-driven fault detection and diagnosis in cooling units using sensor-based machine learning classification, *Sensors* 25 (12) (2025) 3647.
 - [8] S. Potharaju, R.K. Tirandasu, S.N. Tambe, D.B. Jadhav, D.A. Kumar, S.S. Amiripalli, A two-step machine learning approach for predictive maintenance and anomaly detection in environmental sensor systems, *MethodsX*. 14 (2025) 103181.
 - [9] M.K. Farahani, M.H. Yazdi, M. Talaei, A. Rajabi Ghahnavieh, Enhancing energy efficiency in supermarkets: a data-driven approach for fault detection and diagnosis in CO₂ refrigeration systems, *Appl. Energy* 377 (2025) 124479. Part B.
 - [10] R. Wijethunga, H. Nouraei, C. Zych, J. Samarabandu, A. Sadhu, Precision leak detection in supermarket refrigeration systems integrating categorical gradient boosting with advanced thresholding, *Energies*. (Basel) 17 (2024) 736.
 - [11] K. Leerbeck, P. Bacher, C. Heerup, H. Madsen, Grey box modeling of supermarket refrigeration cabinets, *Energy and AI* 11 (2023) 100211.
 - [12] H. Madsen, J. Holst, Estimation of continuous-time models for the heat dynamics of a building, *Energy Build.* 22 (1995) 67–79.
 - [13] N.K. Kristensen, H. Madsen, S.B. Jorgensen, Parameter estimation in stochastic greybox models, *Automatica* 40 (2004) 225–237.
 - [14] MathWorks, Global Optimization Toolbox User's Guide, The MathWorks, Inc., Natick, MA, 2024. Available from, <https://www.mathworks.com/products/global-optimization.html>.
 - [15] M. Suchon, Leveraging machine learning in digital twin models for advanced system identification and predictive maintenance of refrigerator systems. Akademia Górniczo-Hutnicza im. Stanisława Staszica w Krakowie Praca Dyplomowa, 2025, <https://doi.org/10.13140/RG.2.2.23410.00961>.